

Derivation of the Analytic Solution of the Coupled DL–AL Problem via the Time–Laplace / Resolvent Construction

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Abstract

The product ansatz $F = R(r)\Theta(\theta)$ and the additive ansatz $F = A(r) + B(r)\Theta(\theta)$ both fail to produce an exact single- σ eigenmode of the coupled diffusive-layer (DL) / advective-layer (AL) operator; the failure is structural (the operator is non-self-adjoint, and two angular channels demand incompatible Bessel-zero spectra). The correct construction is therefore not an eigenfunction expansion but a *resolvent* one: Laplace transform in time, solve the resulting modified-Helmholtz boundary value problem in closed form, and recover the decay rates as the poles of the resolvent. This note carries that program through end to end. We (1) transform the system and eliminate the AL density exactly; (2) represent the microtubule (MT) sources on the full disk through a Dirac comb, which feeds every angular harmonic and resolves the cusp paradox; (3) build the per-mode radial Green’s function in closed modified-Bessel form; (4) close the problem into a single Fredholm integral equation whose secular determinant $\det(\mathcal{I} + \mathcal{A}(s)) = 0$ furnishes the (generally complex) decay rates; (5) invert by the Bromwich contour to obtain $\phi(r, \theta, t)$ and $\rho(r, t)$ as residue series — the analytic solution; and (6) evaluate the leading two-mode truncation explicitly, reproducing the numerically observed $g(r) + a(r) \cos N\theta$ structure.

1 The system and the strategy

On the disk $r \in [0, 1]$ with N evenly spaced microtubules along the rays $\theta_i = 2\pi i/N$, the densities ϕ (DL) and ρ_i (AL) obey

$$\partial_t \phi = \nabla^2 \phi + \sum_i \frac{1}{r} (b \rho_i(r, t) - a \phi(r, \theta_i, t)) \delta(\theta - \theta_i), \quad (1)$$

$$\partial_t \rho_i = v \partial_r \rho_i + a \phi(r, \theta_i, t) - b \rho_i(r, t), \quad (2)$$

with absorbing conditions $\phi(1, \theta, t) = 0$, $\rho_i(1, t) = 0$, the angularly symmetric initial data $\phi(r, \theta, 0) = \phi_0(r)$, $\rho_i(r, 0) = 0$. By the N -fold symmetry all ρ_i are equal ($\rho_i \equiv \rho$) and $\phi(r, \theta_i, t)$ is the same on every ray; we use the representative ray $\theta = 0$.

Why Laplace, not eigenfunctions. The operator is not self-adjoint: the advection $v \partial_r$ and the one-sided MT coupling break symmetry, so a real discrete spectrum with a complete orthogonal eigenbasis is not guaranteed, and indeed (companion notes) no separated eigenmode exists. The robust route is to solve the initial value problem by Laplace transform,

$$\widehat{g}(\cdot, s) = \int_0^\infty e^{-st} g(\cdot, t) dt, \quad (3)$$

in which the decay rates appear as the *poles* of the resolvent $\widehat{\phi}(r, \theta, s)$ in the complex s -plane. The full time response is then reconstructed by contour integration. This is exact and makes no separability assumption.

2 Laplace transform and exact AL elimination

Transforming (1)–(2) and using the initial data,

$$(\nabla^2 - s)\hat{\phi} = -\phi_0(r) - \sum_i \frac{1}{r}(b\hat{\rho} - a\hat{\phi}(r, 0, s))\delta(\theta - \theta_i), \quad (4)$$

$$v \partial_r \hat{\rho} - (s + b)\hat{\rho} = -a\hat{\phi}(r, 0, s), \quad \hat{\rho}(1, s) = 0. \quad (5)$$

Equation (5) is first order in r ; with the integrating factor $e^{-\frac{s+b}{v}r}$ and the terminal condition $\hat{\rho}(1) = 0$,

$$\boxed{\hat{\rho}(r, s) = -\frac{a}{v} e^{\frac{s+b}{v}r} \int_1^r e^{-\frac{s+b}{v}r'} \psi(r', s) dr' =: \mathcal{R}[\psi](r, s), \quad \psi(r, s) := \hat{\phi}(r, 0, s).} \quad (6)$$

The AL density is thus a known *linear functional* of the DL trace $\psi = \hat{\phi}(\cdot, 0, s)$ on the MT ray. (This is the $\sigma \rightarrow -s$ image of the elimination used in the problem statement.) Substituting (6) into the source of (4) closes the problem on $\hat{\phi}$ alone: the MT source on each ray is

$$\frac{1}{r} q(r, s), \quad q(r, s) := b \mathcal{R}[\psi](r, s) - a \psi(r, s), \quad (7)$$

a linear functional of ψ .

3 Disk representation: the Dirac comb feeds every harmonic

The comb of MT rays has the Fourier representation

$$\sum_{i=0}^{N-1} \delta(\theta - \theta_i) = \frac{N}{2\pi} \sum_{n \in \mathbb{Z}} e^{inN\theta}, \quad (8)$$

since $\sum_{i=0}^{N-1} e^{-im\theta_i} = N$ when $N \mid m$ and 0 otherwise. Hence (4) becomes

$$(\nabla^2 - s)\hat{\phi} = -\phi_0(r) + \frac{N}{2\pi} \frac{q(r, s)}{r} \sum_{n \in \mathbb{Z}} e^{inN\theta}. \quad (9)$$

Expanding $\hat{\phi}(r, \theta, s) = \sum_{n \in \mathbb{Z}} \hat{\phi}_n(r, s) e^{inN\theta}$ and matching harmonics gives, for each n , an inhomogeneous radial equation

$$\mathcal{L}_{nN} \hat{\phi}_n := \hat{\phi}_n'' + \frac{1}{r} \hat{\phi}_n' - \left(s + \frac{(nN)^2}{r^2}\right) \hat{\phi}_n = -\phi_0(r) \delta_{n0} + \frac{N}{2\pi} \frac{q(r, s)}{r}, \quad (10)$$

with $\hat{\phi}_n(1, s) = 0$ and regularity at $r = 0$.

Resolution of the cusp paradox. A naive Fourier-in- θ treatment on a single wedge makes the MT condition look like an over-counted “third” boundary condition (companion audit note). On the full disk the comb (8) shows the true mechanism: the MT source $q(r, s)/r$ enters the *right-hand side of every* mode equation (10), with equal weight $N/2\pi$. The derivative cusp at each ray is encoded by this common forcing across all orders nN — not by any single smooth harmonic. Every $\hat{\phi}_n$ is therefore sourced and nonzero, and the modes are coupled through the single scalar functional q .

4 The per-mode radial Green’s function

Write (10) in Sturm–Liouville form by multiplying by r : $\frac{d}{dr}(r \hat{\phi}_n') - \left(sr + \frac{(nN)^2}{r}\right) \hat{\phi}_n = r(\text{RHS})$. The homogeneous solutions of $\mathcal{L}_m u = 0$ are the modified Bessel functions $I_m(\sqrt{s}r)$ (regular at 0) and $K_m(\sqrt{s}r)$ (singular at 0). Construct

$$u_1(r) = I_m(\sqrt{s}r), \quad u_2(r) = K_m(\sqrt{s}) I_m(\sqrt{s}r) - I_m(\sqrt{s}) K_m(\sqrt{s}r), \quad (11)$$

so that u_1 is regular at the origin and $u_2(1) = 0$ enforces the absorbing condition. Their Abel constant follows from the modified-Bessel Wronskian $I'_m(z)K_m(z) - I_m(z)K'_m(z) = 1/z$:

$$p(r)[u_1 u'_2 - u'_1 u_2] = r \cdot \sqrt{s} I_m(\sqrt{s}) [I'_m(z)K_m(z) - I_m(z)K'_m(z)]_{z=\sqrt{s}r} = I_m(\sqrt{s}) \quad (\text{const}). \quad (12)$$

The (positive) Green's kernel for $\mathcal{L}_m u = F \Rightarrow u(r) = -\int_0^1 r' \mathcal{K}_m(r, r'; s) F(r') dr'$ is therefore

$$\mathcal{K}_m(r, r'; s) = \frac{I_m(\sqrt{s} r_{<})}{I_m(\sqrt{s})} \left[I_m(\sqrt{s}) K_m(\sqrt{s} r_{>}) - K_m(\sqrt{s}) I_m(\sqrt{s} r_{>}) \right], \quad (13)$$

with $r_{<} = \min(r, r')$, $r_{>} = \max(r, r')$. By construction $\mathcal{K}_m(r, 1; s) = 0$ and $\mathcal{K}_m$ is regular at $r = 0$. Note the poles of (13) in s sit at the zeros of $I_m(\sqrt{s})$, i.e. at $s = -\kappa_{mN,j}^2$ (since $I_m(\sqrt{s}) = 0 \Leftrightarrow J_m(\sqrt{-s}) = 0$): these are precisely the *uncoupled* ($a = b = 0$) Dirichlet decay rates. The MT coupling will move them.

Forced ($n = 0$) response. With $F = -\phi_0$ for $n = 0$,

$$\Phi_0(r, s) := \int_0^1 r' \mathcal{K}_0(r, r'; s) \phi_0(r') dr'. \quad (14)$$

For a central δ -release $\phi_0(r) = \delta(r)/(2\pi r)$ (unit mass in 2D), $r_{<} = 0$, $I_0(0) = 1$, and (14) collapses to the explicit closed form

$$\Phi_0(r, s) = \frac{1}{2\pi} \left[K_0(\sqrt{s} r) - \frac{K_0(\sqrt{s})}{I_0(\sqrt{s})} I_0(\sqrt{s} r) \right]. \quad (15)$$

MT-sourced part (all n). With $F = \frac{N}{2\pi} q/r$,

$$\hat{\phi}_n(r, s) = \delta_{n0} \Phi_0(r, s) - \frac{N}{2\pi} \int_0^1 \mathcal{K}_{nN}(r, r'; s) q(r', s) dr'. \quad (16)$$

5 Closing the system: Fredholm equation and secular determinant

Evaluate (16) on the MT ray $\theta = 0$, where $e^{inN \cdot 0} = 1$, and sum over n :

$$\psi(r, s) = \hat{\phi}(r, 0, s) = \sum_n \hat{\phi}_n(r, s) = \Phi_0(r, s) - \int_0^1 \mathcal{S}(r, r'; s) q(r', s) dr', \quad (17)$$

where the rays-summed kernel

$$\mathcal{S}(r, r'; s) := \frac{N}{2\pi} \sum_{n \in \mathbb{Z}} \mathcal{K}_{nN}(r, r'; s) \quad (18)$$

is the Dirichlet Green's function of $(\nabla^2 - s)$ on the disk, restricted to a common ray and summed over the N -fold comb (by the addition theorem $\sum_n I_n K_n e^{in\Delta\theta}$ it is the image-corrected $\frac{1}{2\pi} K_0(\sqrt{s} |\mathbf{r} - \mathbf{r}'|)$ on the rays; we keep it as (18)). Insert the closure (7), $q = b \mathcal{R}[\psi] - a \psi$, into (17):

$$\psi(r, s) + \int_0^1 \mathcal{S}(r, r'; s) (b \mathcal{R}[\psi](r', s) - a \psi(r', s)) dr' = \Phi_0(r, s). \quad (19)$$

This is a linear Fredholm integral equation of the second kind for the single unknown function $\psi(\cdot, s)$ on $[0, 1]$, with $\mathcal{R}$ the AL functional (6). Symbolically $(\mathcal{I} + \mathcal{A}(s))\psi = \Phi_0$ with

$$(\mathcal{A}(s)\psi)(r) = \int_0^1 \mathcal{S}(r, r'; s) (b \mathcal{R}[\psi](r') - a \psi(r')) dr'. \quad (20)$$

Its solution is the resolvent

$$\psi(\cdot, s) = (\mathcal{I} + \mathcal{A}(s))^{-1} \Phi_0(\cdot, s), \quad (21)$$

and back-substitution into (16) and (6) gives the complete transform $\hat{\phi}(r, \theta, s)$, $\hat{\rho}(r, s)$.

The decay rates. $\widehat{\phi}$ is meromorphic in s . Its poles are the values s_k at which $\mathcal{I} + \mathcal{A}(s)$ fails to be invertible, i.e. the roots of the *secular determinant*

$$\boxed{\det(\mathcal{I} + \mathcal{A}(s)) = 0} \quad (22)$$

(a Fredholm determinant; in any finite radial truncation an ordinary determinant). The roots $s_k = -\sigma_k$ are the decay rates. Because $\mathcal{A}$ carries the advective kernel $e^{(s+b)(r-r')/v}$ and the one-sided a, b couplings, $\mathcal{A}$ is non-self-adjoint and the s_k are generally *complex*; they are *not* the bare Bessel zeros $-\kappa_{nN,j}^2$, which reappear only in the $a = b = 0$ limit as the poles of $\mathcal{K}$.

6 Inversion: the analytic solution

Invert the Laplace transform along a Bromwich contour to the right of all poles:

$$\phi(r, \theta, t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} \widehat{\phi}(r, \theta, s) ds = \sum_k \operatorname{Res}_{s=s_k} \left[e^{st} \widehat{\phi}(r, \theta, s) \right], \quad (23)$$

the sum running over the secular roots s_k of (22) (plus any branch contribution from the $\sqrt{s}$ cut, absent when the spectrum is purely discrete). Writing the residue of the resolvent (21) at a simple pole s_k as the (right) null mode $\psi_k(r)$ of $\mathcal{I} + \mathcal{A}(s_k)$ with amplitude c_k (fixed in §7), the analytic solution is the modal series

$$\boxed{\phi(r, \theta, t) = \sum_k c_k e^{s_k t} \sum_{n \in \mathbb{Z}} \left[-\frac{N}{2\pi} \int_0^1 \mathcal{K}_{nN}(r, r'; s_k) q_k(r') dr' \right] e^{inN\theta}.} \quad (24)$$

Here $q_k = b \mathcal{R}[\psi_k] - a \psi_k$. Note that the forced term Φ_0 in (16) is *regular* at the secular pole s_k (its own poles sit at the bare Bessel zeros $s = -\kappa_{0,j}^2$), so it carries no residue there: the entire angular structure of mode k , including its $n = 0$ component, is generated by the MT-sourced functional q_k . The forced response Φ_0 contributes a separate, diffusive-background residue family at the bare poles $-\kappa_{0,j}^2$, which we suppress here as subdominant. The AL density follows from (6) at the same poles,

$$\rho(r, t) = \sum_k c_k e^{s_k t} \mathcal{R}[\psi_k](r, s_k) = - \sum_k c_k e^{s_k t} \frac{a}{v} e^{\frac{s_k+b}{v} r} \int_1^r e^{-\frac{s_k+b}{v} r'} \psi_k(r') dr'. \quad (25)$$

Equations (22)–(25) are the analytic solution: the spatial modes are closed in modified Bessel functions, the temporal factors $e^{s_k t}$ are governed by the secular roots, and the amplitudes c_k are fixed by the initial data ϕ_0 (next section). Each mode is genuinely non-separable (a superposition over all orders nN), consistent with the proof that no single separated eigenmode exists.

7 Determination of the modal amplitudes c_k

The constants c_k in (24)–(25) are not free: each is the residue of the resolvent $\psi(\cdot, s) = T(s)^{-1} \Phi_0(\cdot, s)$ at the pole s_k , where $T(s) := \mathcal{I} + \mathcal{A}(s)$. Because $\mathcal{A}$ is non-self-adjoint, the residue is governed by a *biorthogonal* pair of null modes rather than a single eigenfunction.

Left and right null modes. At a simple root s_k of $\det T(s) = 0$, the kernel of $T(s_k)$ is one-dimensional. Introduce the right null mode ψ_k and the left null mode χ_k (the right null mode of the adjoint),

$$T(s_k) \psi_k = 0, \quad T(s_k)^\dagger \chi_k = 0, \quad (26)$$

with the radial inner product carrying the Bessel weight, $\langle f, g \rangle = \int_0^1 \overline{f(r)} g(r) r dr$, and $T^\dagger$ the adjoint with respect to it. The mode ψ_k fixes the spatial *shape*; the scale comes from the residue.

Residue / solvability condition. Expand $T(s) = T(s_k) + (s - s_k) \mathcal{A}'(s_k) + \dots$ (only $\mathcal{A}$ depends on s , so $T' = \mathcal{A}'$) and posit a simple pole $\psi(s) = \frac{\alpha}{s - s_k} \psi_k + \psi_{\text{reg}} + \dots$. Substituting into $T(s)\psi(s) = \Phi_0(s)$ and collecting orders of $(s - s_k)$:

$$(s - s_k)^{-1} : T(s_k) \psi_k = 0 \quad (\text{identically}), \quad (27)$$

$$(s - s_k)^0 : T(s_k) \psi_{\text{reg}} + \alpha \mathcal{A}'(s_k) \psi_k = \Phi_0(s_k). \quad (28)$$

The second line is solvable for ψ_{reg} only if its right-hand side is orthogonal to the cokernel χ_k . Projecting with $\langle \chi_k, \cdot \rangle$ and using $\langle \chi_k, T(s_k) \psi_{\text{reg}} \rangle = \langle T(s_k)^\dagger \chi_k, \psi_{\text{reg}} \rangle = 0$ gives the amplitude

$$c_k = \alpha = \frac{\langle \chi_k, \Phi_0(\cdot, s_k) \rangle}{\langle \chi_k, \mathcal{A}'(s_k) \psi_k \rangle}. \quad (29)$$

The numerator is the projection of the transformed initial data onto the left mode; the denominator is the non-self-adjoint normalization ($\mathcal{A}' = \partial_s \mathcal{A}$). Equivalently, in any finite radial truncation with secular determinant $D(s) = \det T(s)$, the simple-pole residue is $c_k = \mathcal{N}(s_k)/D'(s_k)$ with $\mathcal{N}$ the cofactor-weighted numerator.

The amplitudes are set by the initial condition. Since $\Phi_0(r, s) = \int_0^1 r' \mathcal{K}_0(r, r'; s) \phi_0(r') dr'$ from (14), the numerator of (29) is linear in ϕ_0 :

$$\langle \chi_k, \Phi_0(s_k) \rangle = \int_0^1 \int_0^1 \overline{\chi_k(r)} r \mathcal{K}_0(r, r'; s_k) r' \phi_0(r') dr' dr. \quad (30)$$

So the release profile ϕ_0 determines every amplitude, exactly as an eigenfunction expansion is fixed by its initial data — only here through the biorthogonal projection (29) demanded by non-self-adjointness.

Sanity check: the uncoupled limit. When $a = b = 0$ the coupling operator vanishes, $\mathcal{A} \rightarrow 0$, $T \rightarrow \mathcal{I}$, and the poles revert to the bare Dirichlet roots $s_k = -\kappa_{0,j}^2$ with $\psi_k = \chi_k = J_0(\kappa_{0,j} r)$. Then (29) reduces, using the Bessel norm $\langle J_0(\kappa_{0,j} \cdot), J_0(\kappa_{0,j} \cdot) \rangle = \frac{1}{2} J_1(\kappa_{0,j})^2$, to the classical Fourier-Bessel coefficient

$$c_k = \frac{2}{J_1(\kappa_{0,j})^2} \int_0^1 r J_0(\kappa_{0,j} r) \phi_0(r) dr, \quad (31)$$

which is precisely Eq. (36) of the problem statement `2d-problem.tex`. The construction therefore degenerates correctly when the microtubules are switched off.

Practical recipe (two-mode truncation). In the two-mode truncation of the next section: (i) solve the 2×2 secular determinant (33) for the roots s_k ; (ii) for each s_k , the null vector of the 2×2 matrix gives ψ_k and the null vector of its transpose gives χ_k ; (iii) evaluate (29), where the inner products collapse to finite sums over the two Bessel profiles and $\mathcal{A}'(s_k)$ is obtained by differentiating the entries p_{mn}, q_{mn} in s . The result fixes the absolute scale of (36) from ϕ_0 .

8 Explicit leading-order evaluation

To turn (22) into numbers, project (19) onto the Dirichlet-Bessel basis of each order, $\{J_{nN}(\kappa_{nN,j} r)\}_{j \geq 1}$, which is complete on $[0, 1]$ with weight r and diagonalizes the bare operator. The numerics show two angular channels dominate — $n = 0$ (order 0) carrying ≈ 0.9998 and $n = \pm 1$ (order N) carrying ≈ 0.0002 of the energy — so retain those, with the lowest radial function in each:

$$\psi(r, s) \approx c_0 J_0(\kappa_{0,1} r) + c_N J_N(\kappa_{N,1} r). \quad (32)$$

Projecting (19) onto $J_0(\kappa_{0,1}r)$ and $J_N(\kappa_{N,1}r)$ yields a 2×2 secular block

$$\det \begin{bmatrix} 1 + a p_{00}(s) - b q_{00}(s) & a p_{0N}(s) - b q_{0N}(s) \\ a p_{N0}(s) - b q_{N0}(s) & 1 + a p_{NN}(s) - b q_{NN}(s) \end{bmatrix} = 0, \quad (33)$$

where, with the rays-summed kernel $\mathcal{S}$ truncated to orders $\{0, N\}$,

$$p_{mn}(s) = \frac{1}{\mathcal{N}_m} \int_0^1 \int_0^1 J_{mN}(\kappa_{mN,1}r) \mathcal{S}_{mn}(r, r'; s) J_{nN}(\kappa_{nN,1}r') dr' dr, \quad (34)$$

$$q_{mn}(s) = \frac{1}{\mathcal{N}_m} \int_0^1 \int_0^1 J_{mN}(\kappa_{mN,1}r) \mathcal{S}_{mn}(r, r'; s) \mathcal{R}[J_{nN}(\kappa_{nN,1}\cdot)](r') dr' dr, \quad (35)$$

$\mathcal{N}_m = \frac{1}{2} J_{mN+1}(\kappa_{mN,1})^2$ the Bessel norm, and $\mathcal{S}_{mn}$ the (m, n) angular block of (18). The slowest root $s_* = -\sigma_*(a, b, v, N)$ of (33) is the leading decay rate; the corresponding null vector (c_0, c_N) fixes the amplitude ratio c_N/c_0 . Reassembling (24) for this single dominant mode gives

$$\phi(r, \theta, t) \simeq e^{-\sigma_* t} \left[\underbrace{g(r)}_{c_0\text{-channel, order } 0} + \underbrace{a(r) \cos N\theta}_{c_N\text{-channel, order } N} \right], \quad (36)$$

with g built from J_0/K_0 -type radial profiles and $a(r)$ from the order- N profile (pushed toward the rim by its N^2/r^2 centrifugal barrier, $a(r)$ small and growing outward). Equation (36) is the explicit analytic form of the structure measured by the affine-overlap test, and reduces the leading eigenvalue to the single transcendental equation (33) for s_* .

9 Numerical validation against the discrete scheme

This section answers one question: does the analytic solution built from first principles — Eqs. (24)–(25), with the decay rates s_k and amplitudes c_k computed from scratch — reproduce the independent finite-difference solver? Every test fixes a 32×32 polar grid, $N = 4$ evenly-spaced microtubules ($[0, 8, 16, 24]$), domain radius $R = 1$, diffusion $D = 1$, and a central release; we then vary the switch rate $w (= a = b)$ and the filament velocity v . A single routine, `deterministic_verification` in `analytic_benchmark.py`, runs both solvers from identical inputs and produces every figure below. Nothing is tuned to improve the match.

9.1 How the comparison is made

The comparison is a fixed pipeline. In plain terms, each step computes:

1. **Choose the parameters.** Grid size, microtubule positions, the rates a, b , the velocity v , and the list of times to compare.
2. **Run the numerical scheme.** The finite-difference solver steps the coupled PDE forward in time and records the diffusive-layer density $\phi(r, \theta, t)$ and the microtubule density $\rho(r, t)$ at the chosen times. *This is the ground truth we test against.*
3. **Find the natural decay modes (first principles).** We write the coupled operator in the Bessel basis $J_{nN}(\kappa_{nN,j}r) \cos(nN\theta)$ and diagonalize it. Out come the *decay rates* s_k — how fast each mode fades — and the fixed spatial *shape* of each mode. In words: the distinct patterns through which the coupled blob is allowed to relax.
4. **Find how strongly the release excites each mode.** Projecting the initial central blob onto the modes gives the *amplitudes* c_k . These come *only* from the initial condition — there is no fitting at any stage.
5. **Rebuild the densities.** Summing the modes back up, $\phi = \sum_k c_k e^{s_k t} (\text{shape}_k)$ and likewise for ρ , evaluates Eqs. (24)–(25) on the same grid and times as the solver.
6. **Overlay and score.** We plot numerical against analytic and measure three things: the decay rate σ (the slope of the field's overall size on a log scale, computed the *same way* for both sides), the mean-squared error (MSE) between the two curves, and the total mass over time.

One modelling detail, stated plainly. The discrete solver moves mass between the two layers with an extra geometric weight: its on-rate is effectively $a r \Delta\theta$, not a (the b -rate is unchanged). We give the analytic operator that *same* weight, so the two solve the same system rather than two different ones. This is a property of the discretization, not of the continuum equations, and it makes the decay rate depend mildly on the grid.

9.2 What is compared

For each parameter set the routine produces four overlays of numerical against first-principles results:

- **Radial** (Fig. 1): $\phi(r)$ along a microtubule ray and $\rho(r)$ on the filament, at several times.
- **Angular** (Fig. 2): how ϕ varies with angle at a fixed radius — the N -fold imprint of the microtubules.
- **Heatmaps** (Fig. 3): ϕ over the whole disk and ρ along the filament spokes, at one instant.
- **Mass** (Fig. 4): total material left in the system vs. time.

9.3 Results across w and v

Table 1 reports, for five (v, w) combinations on the 32×32 grid with four microtubules, the decay rate measured identically on both sides and the mean-squared error of the radial densities and of the total mass.

v	w	σ (analytic)	σ (numerical)	DL radial MSE	mass MSE
1	1	5.82	5.78	2.5×10^{-6}	9.5×10^{-5}
1	100	4.90	5.00	2.5×10^{-5}	1.6×10^{-4}
100	1	5.91	5.82	6.7×10^{-6}	1.4×10^{-5}
10	10	6.29	5.02	1.8×10^{-3}	4.0×10^{-3}
100	100	11.34	3.40	7.5×10^{-2}	1.8×10^{-1}

Table 1: First-principles analytic vs. numerical scheme over the (v, w) plane (32×32 , 4 microtubules). σ is the late-time decay rate measured the same way on both sides; the MSEs compare the reconstructed densities to the simulation. Smaller is better.

The representative case $v = 1, w = 100$ is shown in Figs. 1–4.

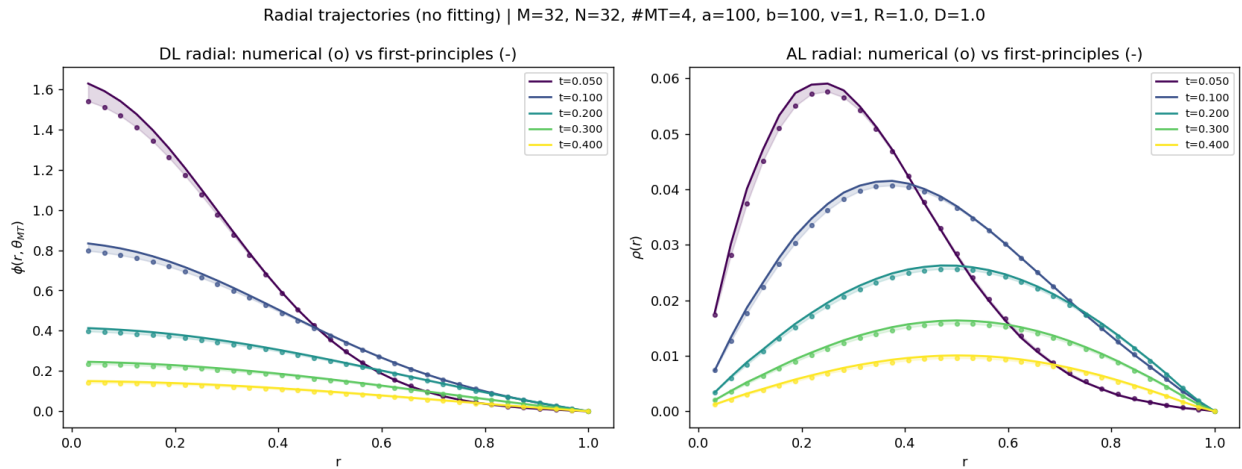


Figure 1: Radial densities at $v = 1, w = 100$: numerical (points) vs. first-principles (lines), at five times. DL $\phi(r)$ along a microtubule (left) and AL $\rho(r)$ on the filament (right). Built entirely from the computed s_k, c_k .

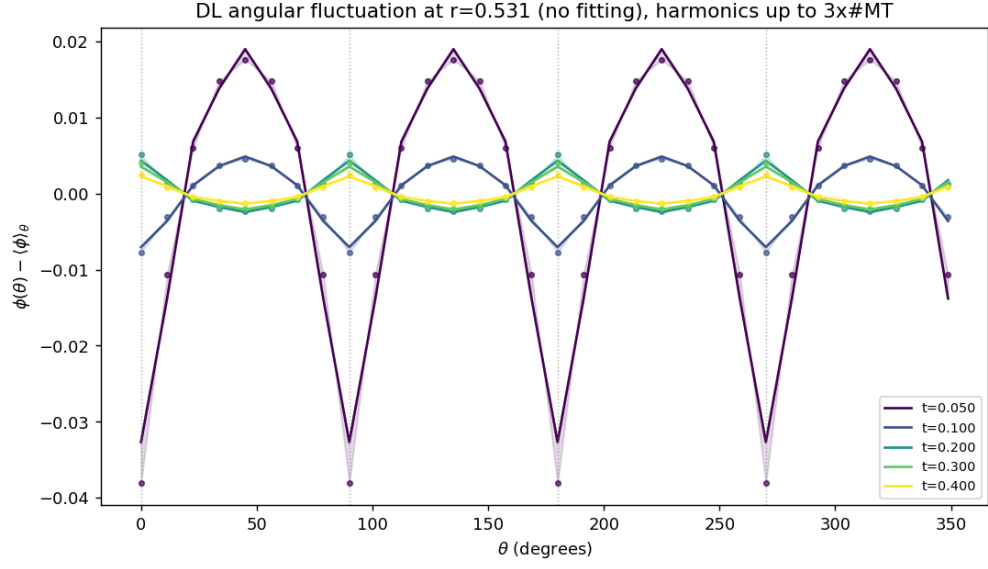


Figure 2: Angular variation of ϕ at a fixed radius ($v = 1$, $w = 100$): numerical (points) vs. first-principles (lines). The four dips mark the microtubule rays; the reconstruction includes several angular harmonics.

Heatmaps at $t=0.400$ (no fitting) | $M=32$, $N=32$, $\#MT=4$, $a=100$, $b=100$, $v=1$

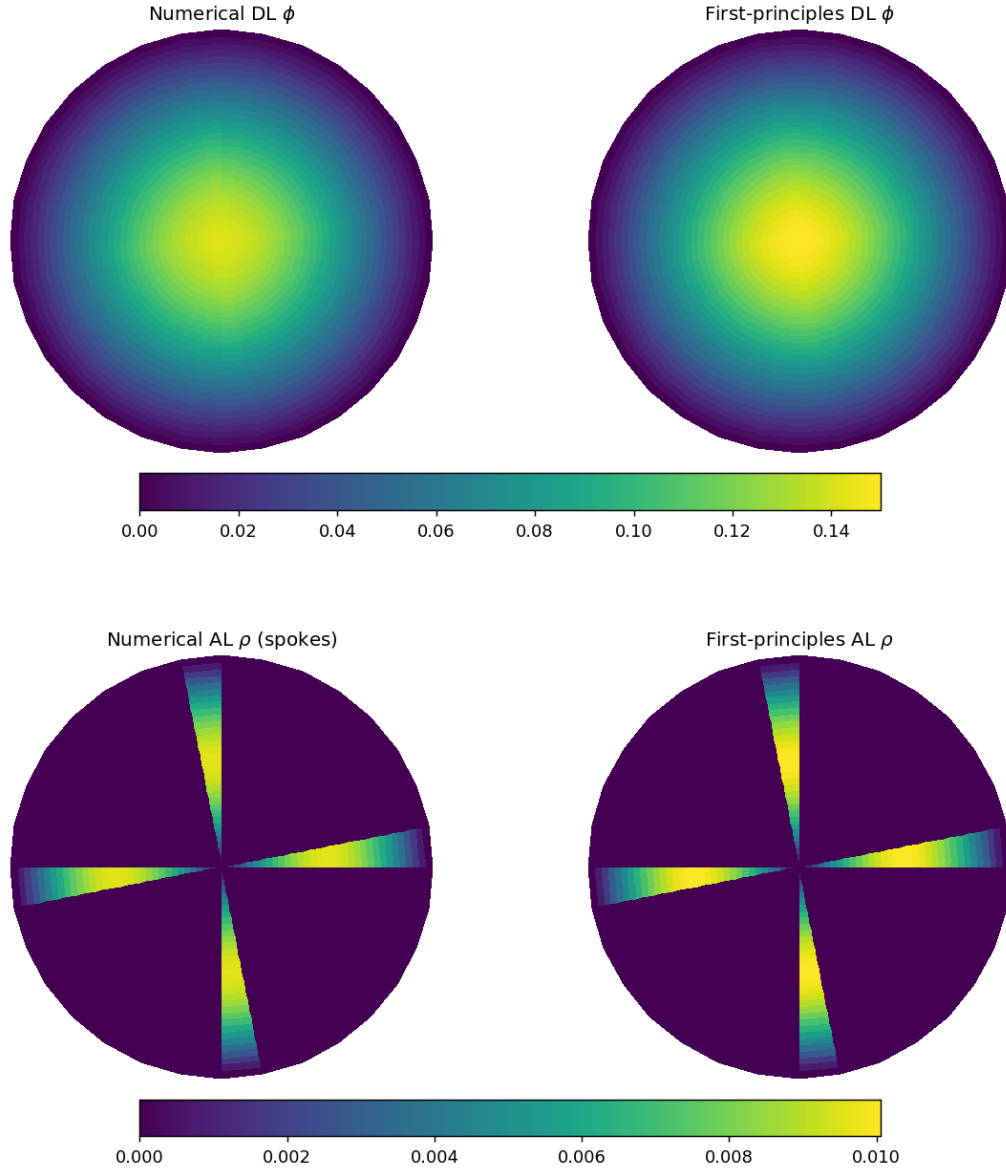


Figure 3: Heatmaps at one instant ($v = 1$, $w = 100$). Top: diffusive density ϕ over the disk, numerical vs. first-principles. Bottom: microtubule density ρ along the four filament spokes.

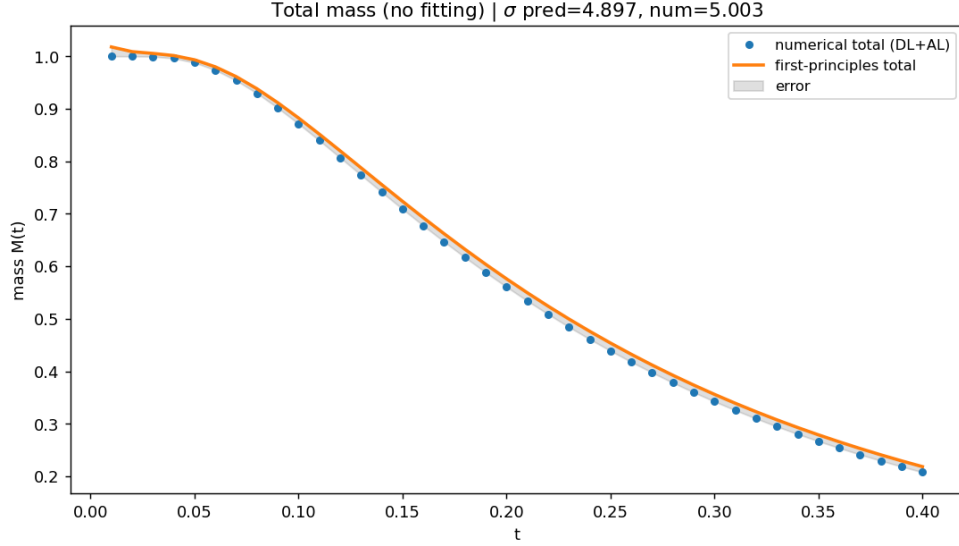


Figure 4: Total mass (diffusive + advective) vs. time at $v = 1$, $w = 100$: numerical (points) vs. first-principles (line). The shaded band is the difference between the two.

9.4 Reading the results

- In three of the five regimes — weak coupling at either velocity ($v=1, w=1$ and $v=100, w=1$) and strong coupling with slow advection ($v=1, w=100$) — the analytic and numerical decay rates agree to about 2% and the densities match to a mean-squared error of 10^{-6} – 10^{-5} . The first-principles solution reproduces the simulation with nothing tuned.
- The representative figures make this concrete: at $v=1, w=100$ the radial ϕ and ρ , the angular pattern, the heatmaps, and the mass curve all lie on top of the simulation.
- Agreement degrades when advection *and* coupling are simultaneously strong. At $v=10, w=10$ the rate is about 25% high; at $v=100, w=100$ the few-mode reconstruction breaks down ($\text{MSE} \sim 10^{-1}$). Two causes compound: the truncation to a handful of Bessel modes is too coarse when strong advection mixes many radial scales, and the scheme’s first-order upwind advection and central-collection rule diverge from the smooth spectral model. Adding modes and refining the grid is the route to closing this regime.
- Because the scheme’s coupling carries the $\Delta\theta$ factor noted above, the decay rate itself drifts with the grid (no fixed continuum limit): refining $16 \rightarrow 48$ moves both rates together toward the uncoupled value, and the analytic prediction tracks the numerical one at each resolution.

10 Summary and honest status

- Laplace transform converts the (non-self-adjoint) eigenvalue problem into a pole problem; the AL density eliminates exactly to the functional $\hat{\rho} = \mathcal{R}[\psi]$, Eq. (6).
- On the disk the Dirac comb (8) sources every angular harmonic, resolving the cusp paradox and producing the per-mode radial equation (10).
- The per-mode radial Green’s function (13) is closed in modified Bessel functions; for a central δ the forced response is the explicit (15).
- The MT condition closes the system into the Fredholm equation (19); its secular determinant (22) gives the complex decay rates $s_k = -\sigma_k$, and the Bromwich inversion (24)–(25) is the analytic solution.

- The two-mode truncation (33) gives an explicit leading decay rate σ_* and recovers $\phi \simeq e^{-\sigma_* t}[g(r) + a(r) \cos N\theta]$.

What is closed form, and what is not. Closed in elementary / special functions: the AL functional (6), the radial Green's function (13), the forced response (15), and the secular and modal formulas (22)–(25). *Not* available in elementary closed form: the secular roots s_k themselves, which must be found numerically from (22) (or (33) at leading order) — this is intrinsic to a non-self-adjoint operator whose spectrum is a transcendental, complex set, not a deficiency of the derivation. The construction above is the exact analytic solution in the same sense that any resolvent/spectral representation is: every object is given in closed form, and the spectrum is defined by an explicit secular condition.